

PalmSim

User Guide

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Preface

Accompanying the release of PalmSim 2.0.0.1 we present this PalmSim user guide. With this guide we hope to (a) help you understand the concept/design of the model, (b) help you become familiar with the use of PalmSim and (c) get you enthusiastic and involved in modelling oil palm. Have fun!

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1 Introduction

In scientific modelling we make a particular part of the world easier to understand and quantify, by referencing it to accepted knowledge and by abstracting away redundant complexity.

PalmSim has grown out of our needs for an intuitive, readable (well-documented and well-implemented) and up-to-date model of oil palm production¹. Our aim is to make understandable and quantify field-level potential and water-limited bunch yield — over time at a monthly time-resolution. To get to this point in a scientific/trust-worthy manner we ask(ed) ourselves: What do we have to describe? How to describe such and such? — given our scientific knowledge. Inspired by PALMSIM (2014) [1] we thus developed PalmSim 2.0.0.1².

In this document we guide you through PalmSim. Primarily by giving an overview of the model at the end of/during which we expect users to get started with examples (available on request) and should be able to resort directly to the very read-able *source code* (also available on request). Next, as a reference we provide a more detailed discourse on model design decisions and the underlying scientific basis of PalmSim.

1.1 On Notation

Throughout this guide we use round brackets () to denote units. Furthermore we use sub-scripts to add information to units. For example, we denote a kilogram dry-matter weight as follows (kg_{DM}). Finally, **bold-face** is used to denote a sub-model and *italic* is used to denote model variables and parameters. For example, the parametrization of the **frond** sub-model involves a *conversion efficiency* ($\text{g}_{\text{DM}}/\text{g}_{\text{assim.}}$) and a variable field *count* of fronds ($1/\text{ha}$).

¹For and overview of earlier oil palm models see the appendix [placeholder].

²By 2.0 we indicate that we made an excursion from PALMSIM for example by implementing the model in Python

2 PalmSim

2.1 Overview



In PalmSim we consider a field of oil palm planted at a certain planting density. One can think of this field as a virtual plantation. The state of the field is modelled by quantifying/tracking the number of fronds, the mass of the fronds, trunks and roots, the flow of assimilates, inflorescences — some of which may have grown out to be bunches — and so on. Furthermore, we consider the site's weather, the site's soil and the management (e.g. ~~trunk~~ pruning). To structure these model elements we have opted for the following hierarchy/collection of **sub-models**:



field

- (a) Weather
- (b) Soil
- (c) Management
- (d) Palm-related
 - i. Fronds
 - ii. Trunk
 - iii. Roots
 - iv. Assimilate pool
 - v. Indeterminate Inflorescences
 - vi. Male Inflorescences
 - vii. Female Inflorescences
 - viii. Bunches

This is meant to express that the **field** model is built by linking a **weather**, **soil**, **management** and **palm** sub-model: The field lends it's properties from the encapsulated sub-models. Similarly, the **palm** sub-model lends it's properties from be encapsulated **fronds**, trunk, etc. sub-models.



To model the dynamics of the field sub-models are coupled, in an intuitive manner. For example, the **fronds** model is coupled to the **weather** model via the solar radiation which conceptually lands on the fronds thus driving gross assimilation. Similarly, the **assimilates** sub-model tracks the assimilates produced by the **fronds**.

To introduce each of these (sub-)models let us briefly specify their primary role, parameters/parametrization and state/rate variables. Let us start with the **weather** and end with the **bunches**.

2.1.1 Weather

The role of the weather model is to keep track of the amount of *solar radiation* /m²/day) and *rainfall* (mm/month) ~~falling on the field~~ — respectively on

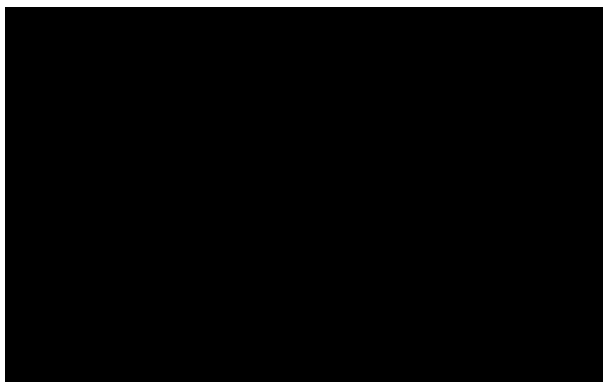


Figure 1: Typical solar radiation(MJ/m²/day) and rainfall (mm/month) time-series for North Sumatra.

the **fronds** and the **soil**. This is done by looking up these values in a simple weather file (a spreadsheet) given the date at which we want to evaluate the model.

The only parameter in the weather model is the *PAR fraction* ³ in the sun-light which by default is set at 0.5 [citation]. This parameter is used to estimate the *PAR* (MJ/m²/day) in the sun-light (which ultimately drives the virtual photo-synthesis).

The weather is modelled by four variables, three of which have already mentioned: (visible?) radiation, rainfall and PAR. Finally we also consider the *number of days with rain* (1/month). This serves as an estimator of the "sun-ness" of the month which is used in/linked to the **soil** model.

2.1.2 Soil

To estimate water-limited yield we estimate the water balance in the soil. That is to say, the role of the soil model is to provide an estimator of plant water availability. This is done in terms of the *soil water deficit* (mm). Here the soil represents the mean soil in the field ⁴. Note, soil modelling is a science in itself so if the following appears complex (at first) do not be alarmed.

The soil's water retention is parametrized by a so-called *water holding capacity* (mm) — an estimator of the amount of water held in the soil after a thorough soaking (compare clay and sand, think porosity, etc.), the *available water capacity* (mm) — an estimator of the maximum amount of water a plant can get out of the "wet" soil and the critical water deficit (mm) — the threshold at which the plant starts to lose function⁵. Finally, to estimate the evapo-transpiration

³photo-synthetically active radiation

⁴For the moment, although possibly interesting to model, we are not considering in soil heterogeneity within a field. Note, we can for the most part deal with such heterogeneity by studying the model for each specific soil

⁵This is linked to some sort of transition. Namely [placeholder]

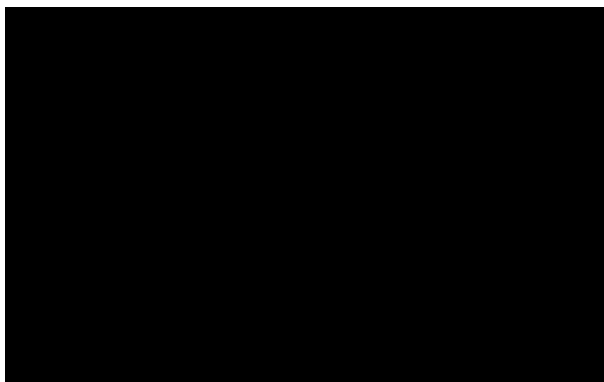


Figure 2: A conceptual model of the soil water balance.

(ET) we consider two parameters: the *high ET* (mm/month) in "sunny" month and the *low ET* in a "rainy" month. Note, the advantage of this method is that it is straightforward and that historically it has been developed specifically for oil palm by IRHO^{6,7}.

The main (state) variable of the soil sub-model, which links to the palm, is the *soil water deficit* (mm) — technically, the soil water deficit relative to/exceeding the critical deficit.

2.1.3 Fronds

Not surprisingly the primary role of the fronds is virtual photo-synthesis: the production of **assimilates** which drives growth and maintenance of the virtual palm. We do so via the *fraction intercepted* (1) light which i.a. depends on the *leaf area* (ha/ha) and the *light-use efficiency* ($g_{\text{assim.}}/\text{MJ}_{\text{PAR}}$). The secondary role is the driving of the inflorescence initiation via the *frond initiation rate* (1/month) through a coupling to the **indeterminate inflorescences** sub-model (since inflorescences initiate in the axil of each new frond).

Due to the different roles (assimilate source, sink, inflorescence initiation) of the fronds sub-model the parametrization is actively involved and can be split in three groups: related to respiration, related to photo-synthesis and growth. In the first group we have the *specific maintenance requirement* ($g_{\text{assim.}}/g_{\text{DM}}/\text{day}$) and *conversion efficiency* ($g_{\text{assim.}}/g_{\text{DM}}$) parameters which respectively help express the cost of maintenance and growth. Concerning photo-synthesis we parametrize the leaf area per frond (m^2) as a function of plantation age (unit's of time after planting), the *light-use efficiency* ($g_{\text{assim.}}/\text{MJ}_{\text{PAR}}$) and a *canopy light absorption coefficient* (1). Concerning growth rate we have

⁶Institut de Recherche pour Les Huiles et Oleagineux

⁷For a discussion see chapter 3, page 58 of [?], the paper [?] or the later more extensive section on the soil model



Figure 3: Frond leaf area (m^2) over time (years after planting) as featured in PalmSim.

parametrized the *potential mass growth rate* ($\text{kg}_{\text{DM}}/\text{palm}/\text{month}$)⁸ and the *leaf initiation rate* ($1/\text{palm}/\text{month}$).

Evidently the main variables are the *frond count* ($1/\text{ha}$), the *frond leaf area* (m^2) and the *frond initiation rate* ($1/\text{palm}/\text{month}$). Note, we also quantify as variables the *mass* ($\text{tonne}_{\text{DM}}/\text{ha}$), *prune rates* ($\text{count}/\text{ha}/\text{month}$), ($\text{tonne}_{\text{DM}}/\text{ha}/\text{month}$).

2.1.4 In general for all organ sub-models

Let us note here that each palm sub-model (fronds, trunk, roots, inflorescences) has a variable *potential mass growth rate* ($\text{kg}_{\text{DM}}/\text{organ}/\text{month}$, via the planting density translatable to $\text{tonne}_{\text{DM}}/\text{ha}/\text{month}$) and thus a corresponding (linked via the conversion efficiency) *potential sink strength* ($\text{tonne}_{\text{assim.}}/\text{ha}/\text{month}$) — in terms of potential assimilate uptake for growth. This potential sink strength is used in the **assimilates** sub-model to determine the actual *assimilate uptake rate* for growth and thus the actual mass growth rate.

Furthermore, there each sub-model has an associated *assimilate uptake rate for maintenance* ($\text{kg}_{\text{a}}/\text{organ}/\text{month}$ or $\text{tonne}_{\text{assim.}}/\text{ha}/\text{month}$).

2.1.5 Assimilates

The assimilates sub-model's role is to book-keep and proportionate the assimilate flows to and from the palm's organs: the **fronds, trunk, roots and inflorescences**. Thus, this sub-model is coupled to each of these sub-models. To repeat the end of the previous section on fronds, each organ sub-model has a variable *potential sink strength* ($\text{tonne}_{\text{assim.}}/\text{ha}/\text{month}$) which is used here to determine the actual sink strength (given the gross assimilation of the fronds).

The only parameter for this sub-model is an index *priority of vegetative growth* (1) which sets in how far vegetative growth gets priority over generative

⁸Note, we convert between mass (growth rate) on a palm (mass/palm) and hectare (mass/hectare) basis via the planting density.

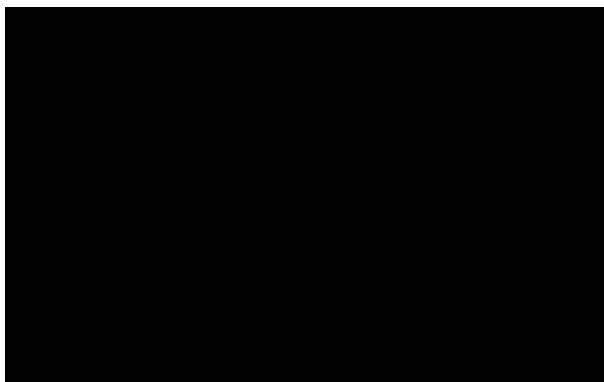


Figure 4: Potential mass growth rates (tonne_{DM}/ha/month) over time (years after planting) as featured in PalmSim for the roots, trunk and fronds.

growth in terms of the partitioned amount of assimilates. This to take into account the general knowledge that oil palms prioritize vegetative growth.

The main variables are *assimilate for maintenance* and *growth* for each organ sub-model.

2.1.6 Trunk

In PalmSim the modelled role of the trunks is to act as assimilate sinks for the **assimilates** sub-model and grow over time. At the end of a plantation life-time the trunks can be considered as harvestable products.

Accordingly the trunk sub-model is parametrized in terms of a *potential mass growth rate* (kg_{DM}/palm/month), the *specific maintenance requirement* (g_{assim.}/g_{DM}/day) and a *conversion efficiency* (g_{assim.}/g_{DM}).

The main variable of the trunk sub-model is it's *mass* (kg_{DM}/palm and tonne_{DM}/ha).

2.1.7 Roots

Similar to the trunk, the modelled role of the roots is to act as an assimilate sink for the **assimilates** sub-model and grow over time.

Accordingly, identically roots sub-model is parametrized in terms of a *potential mass growth rate* (kg_{DM}/palm/month), the *specific maintenance requirement* (g_{assim.}/g_{DM}/day) and a *conversion efficiency* (g_{assim.}/g_{DM}). Furthermore we parametrize root loss/turnover via a constant *mass loss rate* (kg_{DM}/palm/month) and a *mass loss fraction* (1/month).

The main variable of the roots sub-model is it's *mass* (kg_{DM}/palm and tonne_{DM}/ha).

2.1.8 Generative organs — cohorts

~~Now an interesting group of sub-models: inflorescences/generative organs. It turns out~~ that fluctuations in yield can be predominantly attributed to fluctuations in the female to male fraction of inflorescences. Probably this regulation of the *female fraction* (1) has evolved as a means to synchronize/time peak photosynthesis (wet season) with peak fruit growth.

In general the inflorescences are modelled as (monthly) cohorts via a mean inflorescence which has a *mass* ($\text{kg}_{\text{DM}}/\text{inflorescence}$), age (months after initiation),  and the number of inflorescences in the cohort, which we have called the cohort *multiplicity* (1/cohort). This way, abortion/sex differentiation can be taken into account by adjusting multiplicities. For example, if 10% abortion occurs for the cohort in some month we simply adjust the multiplicity accordingly (the aborted fraction is considered to be "metabolically in-active").

Each cohort can be thought of as a small sub-model with it's own sink capacity and potential growth. More about that later. Given the total duration of fruit development from inflorescence initiation a reasonable number (around 60) of cohorts is considered — luckily the computer does so for us.

Now let us introduce the first inflorescence sub-model.

2.1.9 Indeterminate

Each inflorescence cohort starts out as an indeterminate cohort and we thus have an indeterminate sub-model. Each cohort act as a sink, grows and possibly develops — ultimately — into a bunch. Before, fractions of the cohort will virtually differentiate into a **male** and a **female** cohort.

Like the other organ-sub-models, the indeterminate cohort's mean organ is parametrized in terms of a *potential mass growth rate* ($\text{kg}_{\text{DM}}/\text{g}_{\text{ADM}}/\text{month}$) — as a function of months after initiation, a *specific maintenance requirement* ($\text{g}_{\text{assim.}}/\text{g}_{\text{DM}}/\text{day}$) and a *conversion efficiency* ($\text{g}_{\text{assim.}}/\text{g}_{\text{DM}}$). Furthermore, we parametrize the age at which the organ differentiates: *age of differentiation* (months after initiation).

The main variables of each indeterminate sub-model are the *age*, *mass* and the *multiplicity*. Note, since we do not consider abortion before sex differentiation the multiplicity can be considered to be constant.

2.1.10 Male

At the time an **indeterminate** cohort is ought to differentiate we essentially change the sex label of the cohort from "indeterminate" to "male" and the multiplicity  adjusted according to the *female fraction* at differentiation. The ~~mean~~ organ  again modelled in terms of a stalky mass. Male cohorts act as sinks, age, grow and "flower"⁹.

⁹Since we are not modelling for the sake of explaining pollination the male cohorts do not actually flower, they simply abort after anthesis — we hope this does not disappoint you

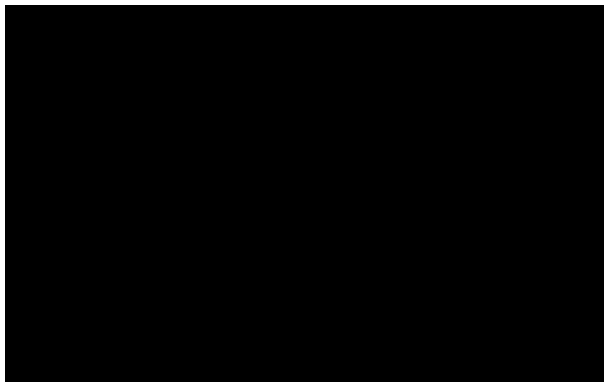


Figure 5: The potential mass growth rate of a stalk ($\text{kg}_{\text{DM}}/\text{ha}/\text{month}$) over time (years after planting) as featured in PalmSim.

Since we are dealing with the same stalky **mean** organ the parametrization of this **mean** organ is identical to that of an indeterminate. The only special parameter considered is the *age of maturation* (months after initiation) which when exceeded allows us to virtually abort the cohort.

The main variables of each male cohort are the *age*, *mass* and the *multiplicity*. Note, since we here as-well do not consider abortion before sex differentiation the multiplicity can be considered to be constant.

2.1.11 Female

Now for something more interesting and variable.

Like the male, at the time an **indeterminate** cohort is ought to differentiate we essentially change the sex label from "indeterminate" to "female" and adjust the multiplicity accordingly. However, furthermore since the female cohorts fruit at some point, past anthesis we consider the representative **mean** organ not only as having a stalky mass but also as having a mesocarp fiber, mesocarp oil and a kernel mass. Female cohorts acts as sinks (relatively strong sinks indeed during bunch setting), age, grow and become a possibly — given sufficient supply of assimilates — heavy oil rich (virtual) bunch.

Since the mean organ is represented by a mass of **stalk**, **mesocarp fibers**, **mesocarp oil**, **kernel** each with an own growth rate, we metrize each of these components by a *potential mass growth rate* ($\text{kg}_{\text{DM}}/\text{pa}$ month) — as a function of months after initiation/anthesis, a *specific maintenance requirement* ($\text{g}_{\text{assim.}}/\text{g}_{\text{DM}}/\text{day}$) and a *conversion efficiency* ($\text{g}_{\text{assim.}}/\text{g}_{\text{DM}}$). Fortunately there is sufficient knowledge of oil palm to do so. Furthermore, since it turns out that (whole) female inflorescences may abort around 1–2 months before anthesis (*floral abortion*) or after anthesis (*bunch failure*) these timings show up as parameters as-well as their typical severity. This involves three parameters per abortion period: *start* of the period, *end* of the period and the typical *abortion*

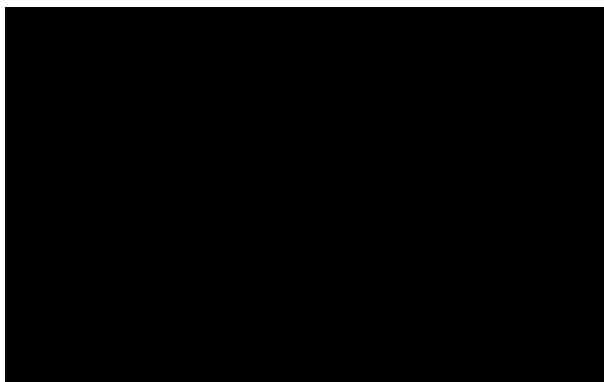


Figure 6: Bunch component mass (kg_{DM}) over time (months after anthesis) as featured in PalmSim.

fraction. Since the severity of the abortion varies with plant stress — which we estimate as a function of drought via the *soil water deficit* (see the **soil**) — there is an additional parameter involved to translate water deficit to additional abortion. Finally, we parametrize the *age of anthesis* and the *age of bunch maturity*.

The main variables of each female cohort are the mass (of all the components) and the multiplicity (bunch count). Since two abortion periods are considered the multiplicity of the cohort is expected drop two times over the course of the cohort life-time.

2.1.12 Management

In the real world and PalmSim plantation management involves in how far the oil palms are pruned. This is addressed in the management sub-model. This sub-model thus couples to the **fronds** sub-model.

The management is parametrized via the desired number of fronds per palm as a function of plantation life-time: the *fronds goal count* (1/palm). Furthermore, here the *planting density* (palm/ha) is stored as a parameter.

The only real variable of the management sub-model is the frond *prune rate* (1/palm/month).

2.1.13 Conclusion

This brings us to the conclusion of the overview of PalmSim. We hope you now have a general feel for the design/working of the model and now expect you to be familiar with the figures shown throughout this section. In case you have not considered to do so please walk through the examples (available upon request) and finally if you are confident we recommend you to dive into the well-documented and read-able source code.

2.2 Details

Here we go over the details of each sub-model: how exactly is the sub-model designed, what exactly are all the variables and parameters? what is the scientific basis, and why did we choose to design the sub-model as such in the first place — and so on.

This sub-section will be released in a next release.

3 Appendix

3.1 Oil Palm Model Comparison

For an overview of release dates see table 3.1. For an overview of release quality see table ??—I tried to be disinterested. For an overview of model features see table 3.1 and 3.1.

name	year	main contributor
OPSIM	1985	Daniel van Kraalingen
SIMPALM	1990	Eric Dufrene
OPRODSIM	2005	Ian Henson
PALMSIM	2010	Alba Castaneda Vera
ECOPALM	2013	Jean-Claude Combres
APSIM	2014	Neil Huth
CLM-Palm	2015	Yuanchao Fan
PalmSim	2017	Willem Hekman



Table 1: An overview of oil palm simulation models. Name: the name of the model. Year: the assumed year of release. Main contributor: the lead author/developer in the main associated paper or release.

name	LUE	SLA	k	diffuse
OPSIM	k	age	planting density, light angle, diffu- siveness	yes
SIMPALM	?	?	?	?
OPRODSIM	?	?	?	?
PALMSIM	constant	constant	constant	no
ECOPALM	SWD	constant	constant	no
APSIM	constant	constant	density	yes
CLM-Palm	?	?	?	?
PalmSim	constant	constant	constant	no

Table 2: An overview of the light-use sub-model features.

name	reproductive growth	floral initiation	floral abortion	bunch failure	female fraction
OPSIM	-	-	-	-	-
SIMPALM	-	-	-	-	-
OPRODSIM	?	?	?	?	?
PALMSIM	SWD	-	-	-	-
ECOPALM	-	IC	IC	IC	IC
APSIM	-	IC	IC	IC	IC
CLM-Palm	-	-	-	-	-
PalmSim	SWD	IC/SWD	IC/SWD	IC/SWD	IC/SWD

Table 3: An overview of the stress-handling features.

3.2 Sensitivity analysis

References

- [1] M.P. Hoffmann, A. Castaneda Vera, M.T. van Wijk, K.E. Giller, T. Oberthür, C. Donough, and A.M. Whitbread. Simulating potential growth and yield of oil palm (*elaeis guineensis*) with palmsim: Model description, evaluation and application. *Agricultural Systems*, 131:110, 2014.

submodule	parameter +5%	ΔY [%]	$\Delta Y_{\max}$ [%]
fronds	specific maintenance	-0.48	-0.45
fronds	specific maintenance	-0.66	-0.62
fronds	conversion efficienc	-0.01	-0.01
fronds	specific leaf area	+1.74	+2.14
fronds	LUE	+6.38	+6.80
fronds	C3	+0.00	+0.00
fronds	k	+1.74	+2.14
fronds	opening a	+0.00	+0.08
fronds	opening b	+0.01	-0.08
fronds	opening c	+0.01	-0.13
roots	specific maintenance	-0.72	-0.72
roots	conversion efficienc	-0.98	-0.96
trunk	specific maintenance	-0.25	-0.21
trunk	conversion efficienc	-0.42	-0.31
trunk	mass loss rate	+0.00	+0.00
indeterminate	abortion fraction	-0.00	-0.00
indeterminate	max new organs	+0.01	-0.26
indeterminate	specific maintenance	+0.00	+0.00
female	abortion fraction be	-0.40	-0.42
female	abortion fraction at	-0.49	-0.53
female	abortion fraction af	+0.00	+0.00
female	conversion efficienc	+4.86	+5.22
female	specific maintenance	-0.59	-0.67
male	abortion fraction	+0.00	+0.00
male	conversion efficienc	-0.02	-0.01
male	specific maintenance	-0.02	-0.01
assim	veg fraction	-7.12	-7.35
assim	fronds fraction	-0.01	-0.01
assim	trunk fraction	-0.42	-0.31
assim	roots fraction	-0.98	-0.96
assim	age of maturity	-0.01	-0.01
assim	max growth female or	+0.01	+0.01

Table 4: Caption